Soyee Lee

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EDUCATION

Chadwick International School, Incheon, South Korea September 2024 – May 2027
IB Diploma Programme, Class of 2027
- Math AA HL; Chemistry HL; Theatre HL; English Language & Literature SL; History SL; Korean A: Language & Literature SL

Mountain Cherry Academy 2021 – 2024
- Principal's Honor Roll (June 2024); Merit Scholarship (2024–25)

ADDITIONAL ACADEMIC PROGRAMS

University of California Scout (Online) May 2026 – August 2026
- AP Biology with Labs Summer Intensive Program (Final Grade A)

Northwestern University Center for Talent Development (Online) July 2024
- Algebra II & Trigonometry Honors

RESEARCH PROJECTS & PUBLICATIONS

Eco-Shield: Natural Edible Coatings to Reduce Fruit Waste | *Team Lead & Co-author* December 2025 – Present
Published, Proceedings of the 16th IEOM International Conference (DOI: 10.46254/AN16.20260220)
- Won the Gold Medal and Champions Award at the 2026 BIEA International STEM Youth Innovation Competition (UK) and Second Place at the 2026 International Conference on Industrial Engineering and Operations Management (presented online).
- Co-authored a study testing three edible coatings against an uncoated control on fruits, measuring spoilage under room-temperature and refrigerated storage over seven days.
- Co-filed Korean [Patent Application No. 10-2026-0198052] (October 2026) for the chitosan–tea polyphenol coating and co-founded and registered FV Lab in Seoul with the two co-inventors to commercialize it; manage its Instagram (@freshveil.lab).

Dietary Management as a Strategy for Reducing Inflammaging in Aging Populations: A Literature Review September 2026
Published, Journal of High School Research, Vol. 3, No. 1 (2026), Article 913260014, pp. 1–12
- Reviewed clinical and observational research linking specific dietary components to inflammatory biomarkers associated with aging, finding that saturated fat, trans fat, and high-glycemic carbohydrates raise these markers while fiber, omega-3s, and MUFAs lower them.
- Highlighted a two-year Mediterranean diet trial showing significant reductions in multiple inflammatory markers, arguing that dietary pattern changes are a practical strategy for managing inflammaging in older adults.

Cognitive Healthspan and the Gut-Brain Axis: A Scoping Review of Korean Traditional Fermented Soy Foods August 2026
- Mapped evidence on Korean jang in relation to gut microbiota and cognitive outcomes, framed by the gap between life expectancy and healthy life expectancy in Korea; manuscript in progress, advised by Dr. Soon-Ah Kang.
- Assessed the diet–microbiota–brain pathway as a preventive strategy for cognitive decline, drawing on the 2023 expansion of the hallmarks of aging to include dysbiosis and chronic inflammation; separated firsthand from secondhand sources and documented contradictory findings to limit confirmation bias.

Okra Mucilage Proportion and Fluorescent Model Particle Capture by Calcium Alginate Beads July 2026
First author; in final review, Journal of Emerging Investigators
- Formulated okra mucilage-calcium alginate composite beads at three mucilage-to-alginate ratios; mean fluorescent model-particle capture in an intestinal simulant (pH 7.4) increased from 61.7 ± 3.4% for alginate-only beads to 87.9 ± 2.5% at the highest mucilage ratio.
- Conducted a sequential oral–gastric–intestinal digestion assay adapted from INFOGEST; cumulative release at the intestinal endpoint was 10.3 ± 1.5% for composite beads versus 22.4 ± 2.6% for alginate-only beads, with results reported descriptively across three trials.

Multidimensional Value Analysis of a Communal Dining Policy: Hyodo Bapsang in a Super-Aged Society June 2026
Featured, Mapo Welfare Foundation Annual Report 2025
- Evaluated Mapo-gu's senior dining program across nutritional, economic, environmental, and sociocultural dimensions; projected an SROI of 15.8x and proposed nationwide adoption.

Fermentation and Aging: Field Observation Report & Proposal May 2026
- Analyzed the 2026 Fermentation Exhibition, examining traditional fermentation techniques, cultural significance, and contemporary preservation practices.
- Submitted a proposal on Korean fermented sauce heritage and food tourism to the Korean Jang and Culture Association, exploring traditional fermentation methods and their contemporary applications.

Comparing the Effects of Aspartame (Artificial Sweetener) and Sucrose January 2026
Published, Journal of Research High School, 2026(6), 443–450
- Analyzed existing literature comparing the two sweeteners' effects on blood glucose response, insulin regulation, the gut microbiome, and long-term health outcomes.

INDIVIDUAL PROJECTS & INVENTIONS

BEACON | *Team Member,* *BEACON Guards* October 2026 – Present

- Designed a system that embeds coded marks in paper or on-screen exam questions so that AI detection apps on smart glasses obscure marked content until the exam closes, without requiring proctors to identify the glasses.
- Built the business case (customer segments, pilot pathway through a Korean education office, per-exam pricing, break-even analysis); entered the Conrad Challenge 2026–27 (Cyber-Technology & Security).

FerroTrace (bladewatch.co.kr) **| *Designer & Sole Inventor*** November 2025 – Present

- Photographed about 50 corroded blades and fracture cross-sections from my club together with a developer under a standardized setup; built a four-level grading rubric based on affected surface area.
- Filed Korean Provisional [Patent Application No. 10-2026-0121155] as sole inventor (July 2026); in September 2026 replaced AI-generated scores with rule-based, in-browser image analysis reporting observations, not diagnoses (coded by the developer).

Lumi's Nutrition Lab: The Junior Scientist Series | *Author & Illustrator* Published Fall 2026

- Wrote and illustrated a planned 10-volume chapter-book series for ages 6–9 that teaches children to observe their own bodies' responses to food; Books 1–3 published in paperback by Baninoon (Kindle self-published); Books 4–10 drafted.
- Piloted the books with six second graders (June 2026) and revised the back matter, adding a glossary, a scientific-method section, and a guide for adults; read to 20 children (October 2026); retained copyright under a June 2025 publishing rights agreement.
- As Design Lead & Content Development Intern at Baninoon (January 2024 – Present), also translated a Korean children's picture book into English and led prop design for a KPIPA-funded children's musical kit.

COMMUNITY LEADERSHIP & SERVICE

Resonance Charity Concert Club (Resonance Ensemble) | *Founder, Violinist & Vocal Soloist* March 2026 – Present

- Recruited a 10-member student ensemble in response to the January 2026 Guryong Village fire that displaced 180+ residents
- Organized two charity concerts, Music Heals, Hope Restores (March 2026) and Harmony for Nepal (October 2026), raising about $500 for fire victims and about $870 so far for Nepal flood relief.

Middle School Teaching Assistant Club (MSTA, LinkED) | *Club Leader* August 2025 – Present

- Led six club tutors in weekly math and Korean tutoring and mentoring for 20 sixth graders; trained members in tutoring and led creation of a study guide.

Hyodo Bapsang ('filial piety table'; free lunch for ages 75+ in Mapo, Seoul) Youth Volunteer Corps
Founder & Leader May 2024 – Present

- Grew the team from 6 to 12 members; served meals on site and organized a lunchtime concert for seniors at the first service.
- Led seven street concerts on Mapo's Red Road to promote the policy and raise funds, helping raise ~$12K cumulative for the Mapo Welfare Foundation's Hyodo Bapsang fund, including $2.3K donated under my own name.

Lee Sang-hee & Friends Orchestra | ***Concertmaster; Honorary Ambassador, NCC*** 2022 – Present

- Performed in nine annual National Cancer Center (NCC) and International Vaccine Institute (IVI) charity concerts, advancing to concertmaster, violin soloist, and concert opening presenter; the orchestra raised ~$486K.
- One of IVI's student Giving Messengers since 2023; as a volunteer intern on IVI's PR & Government Team (June – July 2026), prepared bilingual certificates and letters for 22+ participants of the 24th IVI Charity Concert under a confidentiality protocol.

HONORS & AWARDS

- Gold Medal and Champions Award, BIEA International STEM Youth Innovation Competition (UK) 2026
- Second Place, 16th Annual International Conference on Industrial Engineering and Operations Management (Ankara) 2026
- Two Commendations and a Letter of Appreciation, Mapo-gu Office, for Hyodo Bapsang service 2024–2026
- National Assembly Member's Award at four music competitions, chamber ensemble division 2023–2026
- Commendation of the Prime Minister of the Republic of Korea, awarded to Lee Sang-hee & Friends Orchestra 2025
- Shortlisted Finalist (Law), John Locke Essay Competition 2025
- Commendations, President of the National Cancer Center and Mayor of Goyang 2024
- Encouragement Award, 25th National Middle & High School Volunteer Competition 2023
- Extra: AMC 12 Certificate of Achievement, top 25% (2025); Gold Award, Future Intelligence Students Olympiad, Science and Math (2026); Certificates of Excellence, GeniusCerebrum International Science and Mathematics Olympiads (2026); Top 25% Distinction, New York Times "How-To" Contest (2024–25)

MUSIC, ARTS & ATHLETICS

- **Theater**: Drama Club (October 2024 – Present); cast, Into the Woods (2027, in rehearsal), The Tempest (November 2025), and Anything Goes (February 2025); cast and set designer, Log Out (June 2025; 2nd Place, Incheon Youth Theater Festival).
- **Athletics:** Fencing (Foil): Bronze, 4 competitions including the Seoul Mayor's Cup and KASCA President's Cup (2025); Silver, FILA Cup National Club Championship and Fencing Club Korea Open (2024); Gold, Gyeonggi-do Superintendent of Education Cup (2024).
- **Violin** (chamber ensemble, Song Woo Youth Ensemble): First Place, 2nd & 3rd Young Artist National Orchestra Competitions (online, 2026); National Assembly Member's Award, 7th National Bellas Artes Competition (2025); Grand Prize, 12th Arte Creativa International Competition (2023); First Place, 5th National Bellas Artes Competition (2023).